Smith's Propositions 2 and 4, natively: per curve and per isogeny class

`research/smith_prop24_native_20261014` · 2026-10-14 · protocol `PROTOCOL.md` frozen at commit `31d0beec8` before any study data · run report `reports/run-20261014T1827Z/`

1. Verdicts (fixed wording of the protocol's decision rules)

question	measured	rule (frozen)	verdict
Q1: Smith §8 on random curves, $p = 2^{31} - 1$, $N = 10^6$	0.5019 / 0.5 / 0.501 / 0.1864 (tractable / Prop. 2 / Prop. 4 given Prop. 2 / full_success)	each 98.75% CP interval inside prediction ± 0.01 (0.5016 / 0.5 / 0.5 / 0.1857)	**Smith §8 reproduced (all four fractions equivalent within ±0.01)**
Q2, $p = 7$ (exhaustive)	$D = 0.1758$, $C = 0.4304$, $U = 0.4897$, $r = 0.811$	$r = \frac{C-D}{U-D}$: ≥ 0.75 near the ceiling, ≤ 0.25 near the direct fraction	**near the ceiling**
Q2, $p = 11$ (exhaustive)	$D = 0.1805$, $C = 0.475$, $U = 0.4943$, $r = 0.938$	same	**near the ceiling**
Q2 overall	both primes	common verdict, else prime-dependent	**near the ceiling**
Controls	Example 1: passed · frozen 2026-10-13 counts: reproduced · class-invariant control: closure = direct · independence-null median r: 1 and 1 · point count of X: 18381760 agree, 0 disagree	all pass; instrument invariants all 0	**instrument valid**

Scope. This establishes, for $p = 7$ and $p = 11$ only and for hyperelliptic models $y^2 = f(x)$ with $\deg f = 8$, the share of models whose L-polynomial class contains a model on which Smith's construction gives an $\mathbb{F}_p$-rational isogeny to a trigonal-construction Jacobian. It does not establish that any walk reaches that member, what a walk costs, anything at cryptographic sizes, anything through non-hyperelliptic members of the class, or anything in characteristic 2. Question 1 reproduces Smith's published rates; it is not new evidence about security.

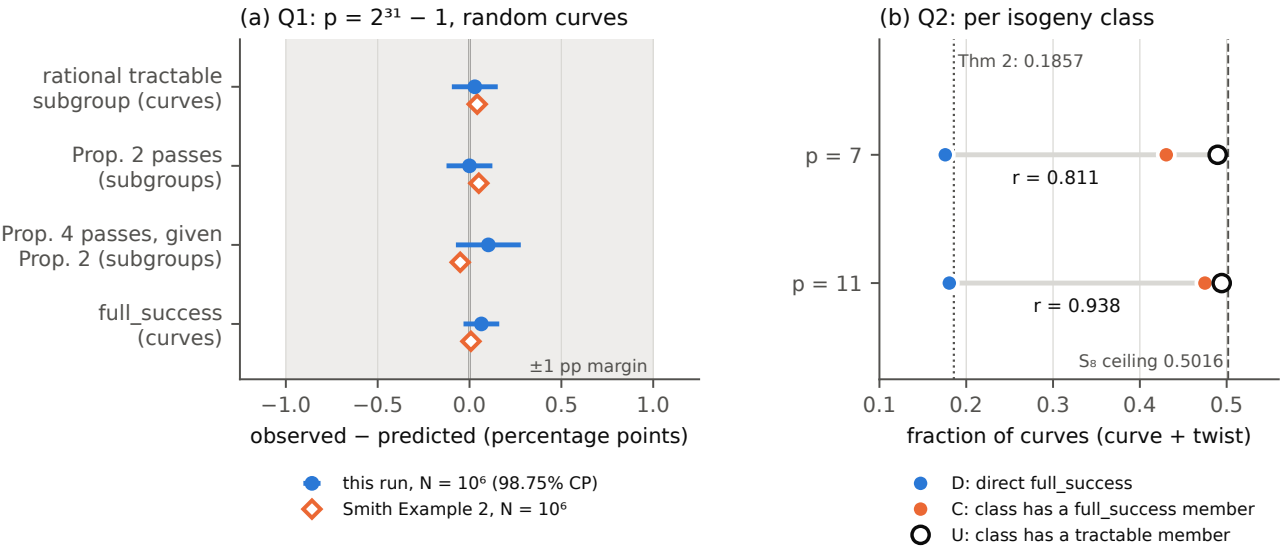


Figure 1: (a) Question 1: observed minus predicted, with the recorded 98.75% Clopper–Pearson interval (blue) and Smith's Example 2 (orange). (b) Question 2, exhaustive census: direct D, class-level C and ceiling U; r is where C sits between D and U. Source `figures/plot.py`, data `data/*_summary.json`.

2. Question 1: Smith §8 reproduced natively

fraction	k / n	value	98.75% CP	cluster bootstrap	prediction (source)	Smith Ex. 2
curves with a rational tractable subgroup	501874 / 1000000	0.50187	[0.50062, 0.50312]	—	0.50159 (Chebotarev density of #S(H) > 0 over S_8 (Lemma 1))	0.50201
tractable subgroups with a rational trigonal map (Prop. 2)	500560 / 1001128	0.5	[0.49875, 0.50124]	[0.49871, 0.50138]	0.5 (Hypothesis 2)	0.50051
of those, subgroups with a rational isogeny (Prop. 4)	250793 / 500560	0.50102	[0.49926, 0.50279]	[0.49913, 0.50286]	0.5 (Hypothesis 3)	0.49949
curves with full_success	186382 / 1000000	0.18638	[0.18541, 0.18736]	—	0.18574 (Theorem 2, eq. (12))	0.18581

Measured. 1000000 curves carry 1001128 rational tractable subgroups (1.0011 per curve; Smith: 1.002244). Every interval contains its prediction and lies inside the ±0.01 margin; Smith's Example 2 value lies in our interval for 4 of 4 fractions. At this p no degenerate case occurred: 1001120 maps, 0 rank deficiencies, 0 zero discriminants, 0 maps outside Smith's normal form, 0 degree drops, and the two rational trigonal maps of a subgroup never disagreed on Prop. 4 (0 cases).

3. Question 2: per isogeny class at $p = 7$ and $p = 11$

quantity (exact unless marked)	p = 7	p = 11
models (curve + twist) / L-classes	1411788 / 5231	35431220 / 21701
U: class has a model with a rational tractable subgroup (= direct)	0.4897	0.4943
D: direct `full_success`	0.1758	0.1805
C: class has a `full_success` model	0.4304	0.475
$r = \frac{C-D}{U-D}$	0.811	0.938
classes with a tractable model / with a `full_success` model	1855 / 1295	7431 / 6221

independence null: C median [2.5%, 97.5%], 199 permutations	0.4897 [0.4897, 0.4897]	0.4943 [0.4943, 0.4943]
measured P_2 (Prop. 2 per subgroup) / P_3 (Prop. 4 given Prop. 2)	0.5594 / 0.4597	0.5408 / 0.4694
Smith's model at this p, heuristic $P_2 = \frac{p+1}{2p}$, $P_3 = \frac{1}{2}$ (heuristic estimate)	0.1924	0.1904
Smith's model at this p, measured $P_2 P_3$ (heuristic estimate)	0.1767	0.1798
asymptotic: Theorem 2 / S_8 ceiling	0.1857 / 0.5016	same

Measured. At both primes C is far above D and close to U: $r = 0.811$ and 0.938, both past the frozen 0.75 threshold, so the verdict is "near the ceiling". C is nevertheless **below** the independence null, which puts C at U in all 199 permutations at both primes: 560 of 1855 tractable classes at $p = 7$ (holding 83610 models) and 1210 of 7431 at $p = 11$ (684915 models) contain no `full_success` model, which independent coins would almost never leave (post hoc, descriptive; Figure 2). Those classes are smaller: median 108 models against 348 at $p = 7$, 420 against 1860 at $p = 11$. From $p = 7$ to $p = 11$, r rises from 0.811 to 0.938 and D moves toward Theorem 2 (0.1758 $\longrightarrow$ 0.1805, against 0.1857); Smith's model with the measured per-subgroup rates gives 0.1767 and 0.1798 for D (heuristic estimate; consistent with his Hypothesis 4 at these primes).

Interpretation (labelled as such). By the frozen rule, the walk extension is worth costing: an isogeny class that holds a model with a rational tractable subgroup usually also holds a model where both of Smith's square conditions pass, so a walk inside the hyperelliptic locus could in principle move 43.04% / 47.5% of models to a `full_success` model, against 17.58% / 18.05% directly. The two conditions are not class invariants (as the tractable subgroup is, by $\chi_\pi \bmod 2$), but they are not independent across class members either: a minority of small classes holds no `full_success` model at all. The reachable ceiling is C, not U. Nothing here prices the walk or builds the isogeny between class members.

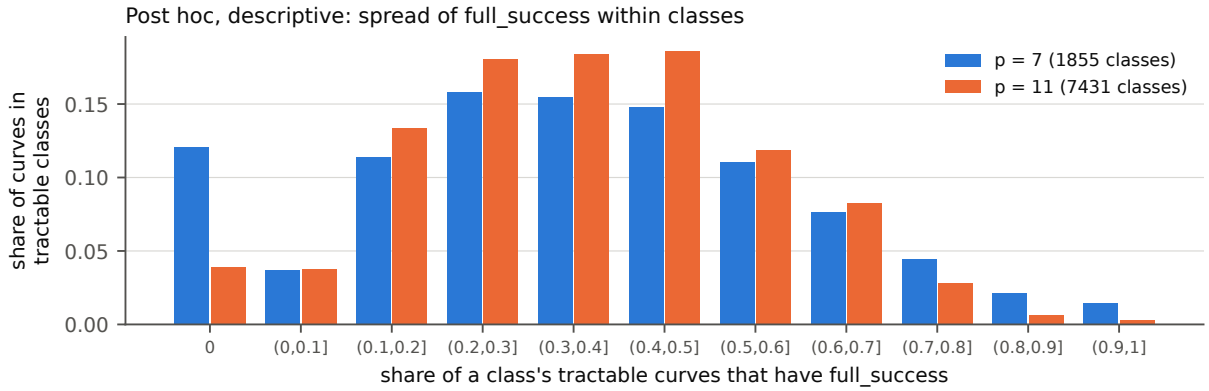


Figure 2: Post hoc, descriptive (protocol amendment 1): share of each tractable class's tractable models that have `full_success`, weighted by models. The bin "0" holds the classes with no `full_success` model. Data `data/describe_classes.json` from `smith_prop24 describe`.

4. Validation

check	result
Smith Example 1 ($\mathbb{F}_{37}$; positive control, same code path)	passed: cycle type 1+1+6, 1 rational tractable subgroup; Smith's $N = x^3 + 16x + 22$, $D = x^2 + 32x + 18$ is one of the 2 computed maps; $f_0, f_1, f_2, \delta_0, \delta_2, \delta_4$ and $64s = \delta_1^2$ equal Smith's; $\#H(\mathbb{F}_{37}) = 42$; Prop. 4 rational; the $x \to 1/x$ model gives the same verdict and its point count of X agrees.
Unit tests (`cargo test --release --lib -- smith_prop24`, 11 tests)	factorisation against `smith_kernel_coverage`; $\#S(H)$ against Lemma 1 and the S_8 action; rational lines meeting the four secants counted by brute force over $\mathbb{P}^3(\mathbb{F}_7)$ = 2/1/0 by the discriminant's class; λ-roots by a generic determinant; normal-form maps exhausted over $\mathbb{F}_7^4$; eq. (7) against the norm; Prop. 4 against the point count of X at $p = 101, 103$; model independence and twist flip; Clopper–Pearson reference values.
Point count of Recillas' curve X against Prop. 4 (every census model and twist)	$p = 7$: 536040 agree, 0 disagree, 0 count errors, 570852 not applicable; $p = 11$: 17845720 / 0 / 0 / 12346260.
Degenerate cases at small p (handled by the frozen rules)	rank $M < 4$: 12570 and 134110 subgroups (Prop. 2 fails); zero discriminant: 77346 and 1357050 no degree-3 map although Prop. 2 passes: 27645 and 535435 subgroups (Prop. 4 fails); maps outside Smith's normal form (norm fallback): 79560 and 1374800. This is why $P_3 < 1/2$ at small p.
Recomputation	`smith_prop24 resummarize --dir data` rebuilds every summary byte for byte from the rows.

5. Claims by class (controlled-experiment skill §1)

- **Exact computation:** every census count, D, C, U, r, the class counts and the Example 1 values.
- **Comparative/statistical:** the four Question 1 equivalences (control: Smith's predictions; 98.75% intervals) and the Question 2 verdicts (frozen thresholds on exact values, gated by controls 4 and 5).
- **Post hoc, descriptive (amendment 1):** "C lies below the independence null" (exact C outside the whole range of 199 stratified permutations) and the within-class spread of Figure 2. Not a registered test.
- **Heuristic estimates:** Theorem 2's 0.1857, the 0.50159 ceiling, and Smith's model at small p.
- **Out of scope / not established:** walk cost, explicit isogenies between class members, cryptographic sizes, characteristic 2, non-hyperelliptic class members.

6. Reproduce

```
cargo build --release --example smith_prop24
B=target/release/examples/smith_prop24; D=research/smith_prop24_native_20261014/data
$B control --out $D
$B random --p 2147483647 --n 1000000 --seed 20261014 --out $D
$B census --p 7 --perms 199 --seed 20261014 --out $D
$B census --p 11 --perms 199 --seed 20261014 --out $D      # about 30 min on 4 vCPU
```

```
$B resummarize --dir $D && $B verdict --dir $D && $B describe --dir $D
python research/smith_prop24_native_20261014/figures/plot.py  # layout only
```

Source. B. Smith, "Isogenies and the discrete logarithm problem in Jacobians of genus 3 hyperelliptic curves", J. Cryptology 22 (2009), arXiv:0806.2995, §3, §5, §6, §8. Prior evidence: `research/weak_isogeny_classes_20261013/`.